

STAA57-Winter 2024: Worksheet-4

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Questions 1:

- i. Suppose X is a random variable, following a $\text{Normal}(\mu = 50, \sigma^2 = 9)$ distribution. Calculate the 1st and 3rd quartile (25th and 75th percentile) of X .
- ii. Find out k which satisfies $P[-k < X < k] = 0.80$ (*hint*: think about the quantile that k corresponds to)

Questions 2:

(The goal is to figure out the distribution of $\bar{X}$ using the example of a fair die)

Task-1:

Write a function in R that mimics the idea of rolling a fair die n times randomly. Let s be the list of those outcomes. Your function needs to return the mean of s .

use this following function and try to write the lines between the curly brackets.

```
sample_die = function(n){  
  
  
}
```

Task-2:

Replicate the function 10000(or a greater number) times and plot a density of the outcomes of the replications.
“code hint: replicate(10000,sample_die(n=2))”

Start with $n = 2$ and gradually change it to 4, 8, 12, 16... and keep drawing the density curve

What difference do you see in the density curve as you increase the value of n .

Instructions for completing this worksheet:

- Create a R markdown file, use your name and student id as the author name.
- Use the command ‘\pagebreak’ between your answers to Q1 and Q2 (this will make sure when you convert to pdf, answer to Q2 starts on a new page)
- Create an html output first, open it in any browser and then print or save as a pdf. (if you already know how to create a pdf directly from R studio then disregard this step)

- Upload your pdf to crowdmark in the place designated for Q1.
- Drag the pages corresponding to Q2 to the place designated to Q2.
- Finish submitting everything by 6pm please.